International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Spring 2025
Course Code: CSE-3529, Course Title: System Analysis and Design
Total marks: 50 Time: 2.5 hours

[Answer the following questions, use separate answer script for Group-A and Group-B.]

GROUP A

			Marks	CO
1.	a)	A university library management system is being developed to handle the borrowing and returning of books. Students can search for available books, borrow them using their student ID, and return them before the due date. The librarian manages the book inventory and student records. When a student borrows a book, the system records the transaction and updates the availability status of the book. If a book is returned late, a fine is calculated by the system. The system also generates monthly reports for the librarian. i) Draw the Level 0 DFD (Context Diagram) for the system. ii) Draw the Level 1 DFD, breaking down the context diagram into at least three sub-processes, such as: • Book Search and Borrowing • Book Return and Fine Calculation • Report Generation	5	CO3

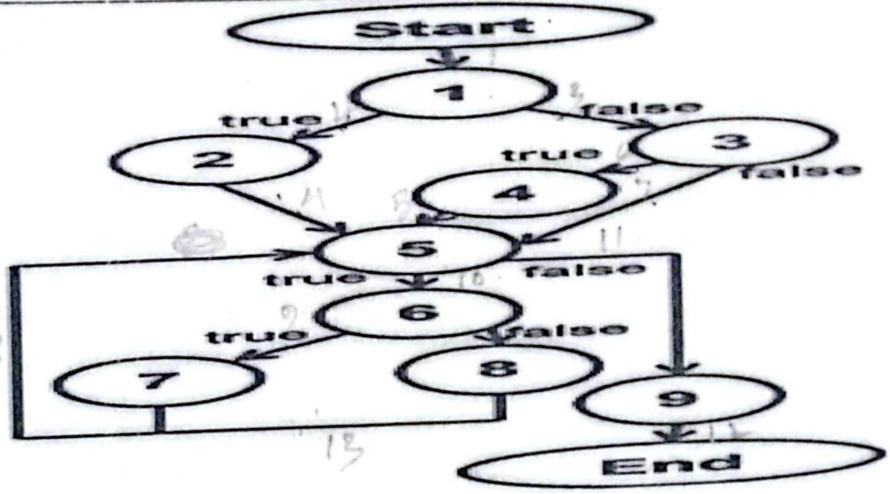
	b)	An online shopping platform wants to determine whether a customer is eligible for free shipping, standard shipping, or no shipping based on the following rules:
		• If the customer is a premium member, they always get free shipping, regardless of the order amount.
		• If the customer is not a premium member, then:
		  o If the order amount is 500 BDT or more, they get standard shipping for free.
		  o If the order amount is below 500 BDT, then:
		    ▪ If the customer is located within the city, they can choose standard shipping with a fee.
		    ▪ If the customer is located outside the city, no shipping is available.
		Draw the Decision Tree and Decision Table for the shipping eligibility scenario.
			5	CO3

		OR		
	a)	"EduPortal" is one of the most widely used online learning platforms in the country. This system bears the following characteristics: i. Students can create a personal profile with required credentials such as name, email, and password to access the platform. ii. Students can browse available courses, enroll in selected courses, and track their progress through their dashboards. iii. Instructors can upload course materials, schedule assignments,	5	CO3

		and view student submissions. iv. Both students and instructors must authenticate themselves before accessing any services. Based on the above scenario, let's identify: a) Find all the external entities b) Design Context Diagram or Level-0 Diagram c) Design Level-1 Diagram using balancing and leveling techniques		
	b)	**A library automation system** determines whether a member can borrow a book based on several conditions. The system evaluates the member's status and the book's availability before making a decision. 1. **If Membership Type is Student, Book is Available, No overdue,** and **Limit not reached** → *Allow Borrowing* 2. **If Membership Type is Guest**, regardless of book status → *Deny Borrowing* 3. **If Book is Reserved** or **Checked Out** → *Put on Hold* (unless Guest) 4. **If Member has Overdue Books** → *Deny Borrowing* 5. **If Borrowing Limit is Reached** → *Deny Borrowing* Based on the above scenario a. Draw a decision tree based on this scenario showing how the system reaches a decision. b. Create a **decision table** listing all combinations of conditions and the resulting decision.	5	CO3

2.	a)	Compare and contrast between top-down and bottom-up system design methodologies with proper examples and appropriate.	5	CO2
	b)	A university wants to design a database for its Online Course Enrollment System. The system must manage information about students, instructors, courses, departments, and enrollments. The system should support the following requirements: a. Students can register and enroll in multiple courses. Each student has a unique ID, name, email, and enrollment date. b. Each course has a unique course code, title, credit hours, and belongs to a specific department. c. Instructors teach one or more courses and are assigned a unique instructor ID, name, email, and designation. d. Each department offers multiple courses and is identified by a unique department ID and name. e. The enrollment of a student in a course records the date of enrollment and the grade received Based on the above scenario i. Identify all entities and their attributes. ii. Determine the relationships between entities and their cardinality. iii. Draw an Entity Relationship Diagram (ERD) representing the scenario.	5	CO4

GROUP B

			Marks	CO
3.	a)	 1. Define cyclomatic complexity and explain its significance in software testing. 2. Calculate the cyclomatic complexity of the above control flow graph.	5	CO3
	b)	Compare and contrast Black Box Testing and White Box Testing in software engineering. a) Define both Black Box Testing and White Box Testing. b) List three key differences between them in terms of purpose, knowledge required, and techniques used. c) For the following scenarios, state whether Black Box Testing or White Box Testing is more appropriate. Justify your answer: i) Verifying the output of a login system without looking at the internal code ii) Testing all possible paths in a sorting algorithm's implementation	5	CO2
4.	a)	Explain the process of software system implementation in the software development life cycle. What are the key activities involved in the implementation phases .	4	CO2
	b)	A city hospital uses a Hospital Management System (HMS) that handles patient records, appointments, billing, and staff scheduling. Over time, the system requires regular maintenance to ensure smooth operation and data security. The hospital's IT department is responsible for maintaining the system.The following maintenance activities are required: Fixing software bugs, corrupted data, or unexpected system crashes.Updating the system to work with new operating systems, hardware, or medical devices. Enhancing performance, adding user-friendly features, or improving the interface.Regular backups, security updates, and optimization to prevent future issues. a) Identify types of maintenance activities required for the above Hospital Management System and describe each types with respect to above scenario. b) Mention the factors that influence system maintainability	6	CO3

5.	a)	A customer uses an online banking system to manage her accounts, transfer funds, and pay bills. The bank ensures that only authorized users can access their accounts by enforcing strong password policies and two-factor authentication. One day, the customer notices that a recent transaction shows a wrong amount — she transferred 5,000 BDT, but the system recorded 50,000 BDT. She reports this issue to the bank immediately. The bank's technical team investigates and finds that a data corruption occurred during a system update. Later that day, several users report that they are unable to access their accounts for several hours due to a denial-of-service attack on the bank's servers. • Identify how the scenario reflects the three core security goals. Explain each security goal with reference to the scenario. • Suggest one possible solution or prevention method for each of the three goals to avoid such issues in the future.	5	CO2
	b)	What is security threat? Demonstrated different types of security threats with proper example.	5	CO1
		OR		
	a)	What are the *security issues* in the context of information systems? Identify and explain **four common security issues in a computer system**	5	CO2
	b)	Explain the concept of system vulnerabilities in computing. Discuss how vulnerabilities can affect each of the following system resources with one example for each: Hardware, Software, Data.	5	CO1

Exposure, Vounarabi lity, Threat, Controls

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Spring-2025

Course Code: CSE 3521 Course Title: Computer Architecture

Total marks: **50** Time: **2.5 hours**

	Group-A	M	CO
1. a)	Suppose you are given an instruction add \$s1,\$t3,\$t4. Describe the complete operation of the datapath using the instruction with a figure.	5	3
b)	Design a multi-cycle data path for handling basic instruction of MIPS.	5	3
2. a)	How does the execution of different instruction types differ in multi-cycle datapath. Show the steps.	5	3
b)	An instruction set consists of the following instruction mix: 35% ALU instructions; 20% Load instructions; 15% Store instructions; 25% Branch instructions; 5% Jump instructions 1. Calculate the overall CPI for this instruction mix in a variable-cycle implementation. 2. If the processor is redesigned with a fixed-cycle, where every instruction takes 3 cycles, what is the new CPI? 3. Compare the CPI in both cases and discuss the trade-offs between using a variable-length cycle design versus a fixed-length cycle design.	5	2
	OR		
2. a)	What changes are required for the multiplexer in multicycle Datapath?	5	3
b)	A multicycle CPU has a **clock time of 250 ps** and the following instruction mix: Instruction: R-type — 60% — 4 cycles; lw — 40% — 5 cycles The program has **100,000 instructions**. **Q:** What is the total CPU time?	5	2
	Group-B		
3.a)	Explain the concept of pipeline stalls caused by data hazards. Propose two techniques to resolve or minimize data hazards.	5	3
b)	Draw and explain the pipeline stages involved in executing a store instruction in a pipelined architecture. Discuss how hazards are handled during execution.	5	3
	OR		
3.a)	Compare and contrast pipelined and single-cycle instruction execution.Use timing diagrams and MIPS instruction examples to support your explanation. Include assumptions about stage delays and overall speedup.	5	3

b)	Design a pipeline schedule for the following MIPS instructions to minimize hazards:
	lw \$t1, 0(\$t2)
	add \$t3, \$t1, \$t4
	sub \$t5, \$t3, \$t6
	a) Identify the types of data hazards present.
	b) Suggest how forwarding or stalling could resolve them.
	c) Reorder the instructions (if possible) to minimize stalls.
		5	3

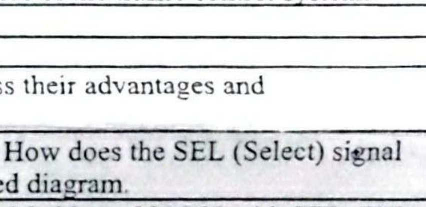
4.a)	Discuss the working of a **direct-mapped cache**. Include the addressing scheme and explain how hits and misses are determined.	5	2
b)	What is virtual memory? How does the **page table** help translate virtual addresses into physical addresses?	5	2
5. a)	Describe how Translation Lookaside Buffer **(TLB)** improves memory access time. What happens on a TLB miss?	5	2
b)	What is **handshaking** in asynchronous communication? Describe its steps with an example of a read request from an I/O device.	5	2

Table in 2. b):

Instruction	%	Cycles
R-type	60	4
lw	40	5

International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE Semester Final Examination, Spring-2025
Course Code: CSE-3523 **Course Title:** Microprocessors, Microcontrollers and Embedded Systems
Time: 2 hours 30 minutes **Total Marks: 50**

	Group-A	M	C
1a	Differentiate between maskable and non-maskable interrupts in 8086 with real-life examples.	3	2
b	Explain how a fixed interrupt vector can be generated using a DIP switch and 74ALS244 buffer during INTR in 8086. Include a neat and labeled circuit diagram.	4	2
c	Describe the sequence of operations performed by 8086 when a divide-by-zero operation occurs. Include the register changes and use of the stack during Type 0 interrupt.	3	2
	OR		
1. a	Suppose an external device generates an interrupt with vector number 60H. Describe how the CPU handles this interrupt after receiving the request. Including the steps involved in locating and executing the corresponding ISR.	5	2
b	Discuss the significance of the lower 3 interrupt types (Type 0–2) in 8086. Mention how each is triggered and handled.	5	2
2. Read the passage and answer the questions 2(a–c): Design a smart traffic signal system for a city that processes sensor inputs, prioritizes emergency vehicles, and reduces congestion using an embedded unit with multitasking, real-time interrupts, and optimized processing.			
a	Identify and justify 3 key processor features for the system. How does dual-core architecture improve multitasking?	3	2
b	Based on this scenario, suggest which type of processor (e.g., embedded, dual-core, arithmetic, bit-slice) would best suit the system. Explain your choice with architectural and application justification.	4	2
c	Explain how the use of cache memory and pipelining can enhance the performance of the traffic control system.	3	2

	Group-B		
3. a	What are the differences between isolated I/O and memory-mapped I/O? Discuss their advantages and disadvantages with relevant instruction examples.	3	3
b	Describe the internal operation of a basic input interface using a tri-state buffer. How does the SEL (Select) signal help in transferring data from input switches to the CPU? Illustrate with a labeled diagram.	3	3
c	What is handshaking in microprocessor I/O systems? Explain Mode 1 input handshaking of 82C55 with STB, IBF, INTR, and RD signals, including a timing diagram.	4	3
4a	Differentiate microprocessor and microcontroller by architecture, functionality, and application, with one example each.	3	3
b	8051 microcontroller comes in a 40-pin DIP package. Draw and explain the pin diagram of the 8051, identifying the purpose of the following pins: 1.EA (External Access) 2. PSEN (Program Store Enable) 3.ALE (Address Latch Enable) 4.RST (Reset) 5. Port 0 and Port 3 special functions.	4	3
c	Write an 8051 assembly language program to multiply two numbers stored at memory locations 40H and 41H. Store the result in registers A and B.	3	3
	OR		
4a	Write a program to add the values stored at memory locations 40H and 41H, and store the result at 50H and 51H.	4	3
b	Explain how a DC motor can be controlled using pin P1.3 and P1.4 of the 8051 microcontroller. Why is L293D necessary, and how do direction control signals work? Include circuit and code	6	3
5.	Look at the figure of FSM(Seat Belt Warning System) at right and answer questions (a-c): 		
a	Identify all states and events in the FSM. Explain each state in the context of a car's seat belt warning system, complete missing transitions, and explain the FSM.	3	3
b	Describe two conditions triggering transition from the 'Waiting' state and their effects. Provide another FSM example, e.g., a traffic light controller.	2	3
c	Identify three limitations of traditional embedded design and their impact on performance. Explain co-design, its benefits, and give an RTOS-based example.	5	3

International Islamic University Chittagong

Department of Computer Science and Engineering (CSE)

B.Sc. Engineering in CSE

Final Examination, Spring 2025

Course Code: EEE-2421 | **Course Title:** Electrical Drives and Instrumentation

Time: 2.5 hours | **Full Marks:** 50

Answer all the questions. The figures in the right-hand margin indicate full marks

Course Outcome (COs) and Bloom's Level (BL) are mentioned in additional Columns

Group-A

1 (a)	Explain the working principle of a Variable Frequency Drive using a block diagram.	CO2	C	5
1 (b)	A 4-pole, 3-phase induction motor operates from a supply with a frequency of 50 Hz. Calculate: i. The speed at which the magnetic field of the stator is rotating. ii. The speed of the motor when the slip is 3%. iii. The speed of the rotor when the slip is 0.04. iv. The frequency of the rotor currents when the slip is 0.03. v. The frequency of the rotor currents at standstill.	CO3	E	5
	Or			
1 (a)	Briefly explain the working principle of three phase induction motor. Discuss how Variable Frequency Drive (VFD) controls the speed of an induction motor.	CO1	U	5
1 (b)	A 3-φ induction motor is wound for 5 poles and is supplied from 50-Hz system. Calculate **(i)** the synchronous speed **(ii)** the rotor speed when slip is 5% and **(iii)** rotor frequency when rotor runs at 500 rpm.	CO1	A	5
2 (a)	Suppose you are working in an industry. You need a motor with constant speed and that can operate in lagging to leading power factor for some applications. Select an appropriate motor for your desired application, explain the working principle of your selected motor, and the reasons for your selection.	CO2	C	5
2 (b)	Briefly analyze the operation of the variable reluctance stepper motor and the permanent magnet stepper motor, and compare them.	CO3	An	5

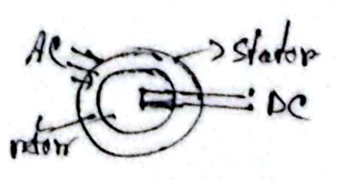
Sync Motor

Group-B

3 (a)	Discuss the three stages of the generalized measurement system and types of errors in the measurement system.	CO3	U	5
3 (b)	For the following given data, $X_1 = 43.7$, $X_2 = 49.1$, $X_3 = 51.2$, $X_4 = 46.6$, $X_5 = 44.7$, Calculate: a) Arithmetic mean b) Deviation of each value c) Algebraic sum of the deviations d) The average deviation e) The standard deviation	CO3	E	5

$N_s = \frac{120 f_s}{P}$

$f_r = s f_s$



$\dot{E} = C(T_2 - T_1) + B(T_2^2 - T_1^2)$

Or

3 (a)	Define the terms Accuracy and Precision. Also, describe the complete process of measurement by explaining the steps taken **before, during,** and **after** the measurement.	CO2	R	5
3 (b)	The following 10 voltage readings were recorded during an experiment: **54.58, 54.73, 54.86, 54.95, 55.00, 55.00, 55.08, 55.19, 55.31, 55.30** volts. Calculate the following: i) Arithmetic Mean **ii)** Deviation of each value **iii)** Average Deviation **iv)** Standard Deviation v) Variance	CO2	An	5
4 (a)	Sketch the three-op-amp instrumentation amplifier and derive the expression for the overall gain. Also, mention the advantages of using this configuration.	CO2	Ap	5
4 (b)	Explain with the help of a functional block diagram, the principle of operation of a digital frequency meter.	CO3	C	5
5 (a)	**I.** A displacement transducer with a shaft stroke of 3 inch is applied to the circuit of Figure 1. The total resistance of the potentiometer is 5kΩ, and the applied voltage $V_T = 5V$. When the wiper is 0.9 inch from B, what is the value of the output voltage V_O?	CO3	E	5

RDT

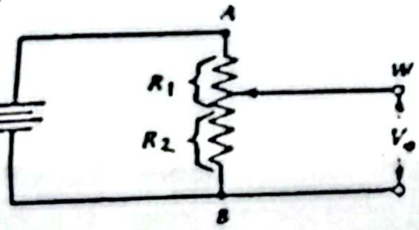


Figure 1.

II. During experiments with a copper-constantan thermocouple, it was found that $c = 3.75 \times 10^{-2}$ mV/°C and $k = 4.5 \times 10^{-5}$ mV/°C². If $T_1 = 1000$ °C and the cold junction T_2 is kept in ice, compute the resultant electromotive force.

5 (b)	What is a transducer? Discuss the classification of transducers and prove the gauge factor equation for a strain gauge.	CO2	U	5

International Islamic University Chittagong

Morality Development Program (MDP)

Semester End Examination, Spring– 2025

Course Title: Concepts of Moral Development

Course Code: MDP-3503, 5th Semester

Full Marks : 50 **Time: 2.50 Hours**

[Figures on the right hand margin indicate full marks]

Answer the following questions

Q.1

a) What is your understanding of an ideal society? 2

b) On what ground students are considered as a very crucial community of a society? 3

c) In what ways students can play a pivotal role in the formation of an ideal society? 5

Q.2

a) Define 'Dowry' and 'Mahr' 2

b) Elucidate the negative impact of dowry in a society 4

c) How the students can contribute in the development of a dowry-free society? 4

Q.3

a) Define 'Entrepreneurship'. 2

b) Figure out the features of good entrepreneurship? 3

c) Explain the importance of Halal earnings from Islamic perspective 5

or

a) What are the misconceptions about Islam?

b) In what means misconceptions are being created against Islam?

c) How to remove the all-pervasive misconceptions regarding Islam?

Q.4

a) Explain the Islamic & Western concepts of entertainment. 3

b) What types of entertainments are allowed in Islam? 3

c) Mention Shariah ruling on alcohol, pornography, extramarital relation and living-together. 4

or

a) In what ways people fall in the gambling-trap? 4

b) Write a note on the far reaching social disaster caused by gambling in the society. 6

Q.5

a) Define Human rights in Islamic perspective. 2

b) Describe the rights of minorities and other religious groups in an Islamic state. 4

c) Narrate the rights of relatives and neighbors in the light of Islam. 4

International Islamic University Chittagong
Center for General Education
Semester End Exam- Spring 2025

Course Code: URED- 3503 (URED- 3101 for LLB)
Course Title: Political Thoughts and Social Behavior

Time: 2.5 Hours **Full Marks: 50**

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

SL	Questions	Marks	CLO	Bloom's taxonomy domain
1	Critically evaluate how marriage serves both as a social contract and a spiritual bond in Islam. **Or,** Analyze the wisdom behind the categories of prohibited marriages (*Maharim*) in Islamic law	10	3	Evaluate & Analyze
2	"Our Lord! Grant us from among our spouses and off springs comfort to our eyes and make us an example for the righteous"- based on this verse; explain the prescribed duties and obligations of a Muslim Spouse.	10	3	Create
3	Analyze how Islamic guidelines on dress reflect the broader values of modesty, identity and social ethics.	10	3 Prohibition of Riba Law of inheritance	Analyze
4	"The Islamic economic system ensures justice and equity while discouraging exploitation and wealth accumulation among the rich persons"- justify the statement	10	3	Evaluate
5	"All things are lawful for mankind except what have been prohibited explicitly by Allah and his messenger"- explain the statement.	10	3	Create

==========

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE, Final Examination, Spring 2025

Course Code: CSE-3527 **Course Title:** Compiler

Total Marks: 50 **Time:** 2 Hours 30 Minutes

[Answer *all* the following questions. Figures in the right hand margin indicate full marks. Use a Separate answer script for Group-A and Group-B.]

Group A

Marks COs

1\. a) The following is a grammar for regular expressions over symbols a and b only, using + in place of | for union, to avoid conflict with the use of vertical bar as a meta symbol in grammars: 5 3

rexpr → rexpr + rterm | rterm
rterm → rterm rfactor | rfactor
rfactor → rfactor * | rprimary
rprimary → a | b

a) Left factor this grammar.
b) Does left factoring make the grammar suitable for top-down parsing?
c) In addition to left factoring, eliminate left recursion from the original grammar.
d) Is the resulting grammar suitable for top-down parsing?

b) Compute FIRST and FOLLOW for the following grammar: 5 3

E → T E′
E′ → + T E′| Ɛ
T → F T′
T′ → *F T′| Ɛ
F → (E) | id

2\. a) Consider the following grammar: 5 3

S → AA
A → aA | b

i. Construct the augmented grammar for the given grammar.
ii. Derive the SLR (1) items for the grammar and then construct the parsing table.

b) For the grammar S → 0 S 1 | 01, indicate the handle of string "000111" using Shift Reduce parsing. 5 3

OR

2\. a) Draw a hierarchy diagram to differentiate the all parsers in compiler construction and also draw the block diagram of LR parser. 2 3

b) Consider the following grammar G_1 over the alphabet $\Sigma = \{a, b, c, d\}$. You want to use an SLR(1) or CLR(1) parser for the strings defined by G_1. You only need to provide accepting states, including their items, and transitions between accepting states. 8 3

$S \to aBa \mid AB \mid d$
$A \to b \mid c$
$B \to a$

i) Draw the DFA transitions and state items of the LR(0) item collections. The first LR(0) DFA state item (I_0) already been provided in **Fig. 2** for your reference.
ii) Construct SLR(1) parsing table from LR(0) DFA state items. The parser table includes both the action and goto sections.
iii) Is the grammar SLR? Justify your answer.
iv) Show the stack implementation along with the output parse tree for the input string "**ca**".

I_0:	$S' \to .S$
	$S \to .aSa$
	$S \to .AB$
	$S \to .d$
	$A \to .b$
	$A \to .c$

Fig. 2

[redacted]

Group B

3. a) A Syntax-directed definition of a simple desk calculator is given below: 5 3

Sl. No.	PRODUCTION	SEMANTIC RULES
1.	$L \to E\ n$	L.val = E.val
2.	$E \to E_1 + T$	E.val = T. val + T. val
3.	$E \to T$	E.val = T. val
4.	$T \to T_1 * F$	T. val = T_1.val × F. val
5.	$T \to F$	T.val = F.val
6.	$F \to (E)$	F.val = E.val
7.	$F \to$ digit	F.val = digit.lexval

Give annotated parse tree and dependency graph for the above expression:
(3 + 4) + (5 + 6) n.

b) Step by step construct syntax tree using the following grammar and string " a-4-c": 5 3

$E \to E_1 + T$

$E \to E_1 - T$

$E \to T$

$T \to (E)$

$T \to \textbf{id}$

$T \to \textbf{num}$

OR

3. a) Syntax directed translation: 2+2+1 3
Write the functions to construct the syntax tree for arithmetic expression. Consider the following logical operations CFG G_0 over the language Σ = {`not`, `or`, `and`, `true`, `false`} with every non-terminal has a synthesized attribute val of Boolean type:

B → B and C | B or C | C

C → *not* C | *true* | *false*

Give the syntax-directed definition of the following grammar G_3. Draw the annotated parse for the input expression string "**not true or false and true**"

b) Type checking: 1+2+2 3

i) Write the purposes of the type-checker on compiler?

ii) Explain with the semantic rules of type checking for C programming expressions. statements and function in compiler type checker.

iii) Write semantic rules for the following type checking expressions:

$E \to E_1\ op\ E_2$

$E \to E_1 [\ E_2]$

Run-time environment: 5 4

i) What is the activation tree in the run-time environment of a compiler for a function?

ii) Draw the activation tree for the following c programming function to generate the n^{th} Fibonacci number, if n=7:

```
int Fibonacci(int n){
    //Base case: if n is 0 or 1, return n
    if (n <= 1){
        return n;
    }
    return Fibonacci(n - 1) + Fibonacci(n - 2);
}
```

iii) Explain the memory allocation strategies for the run-time environment.

Intermediate code generation: 5 5

Translate the arithmetic expression '$x + x*(y - z) + (y - z)*w$" into:

i) Syntax tree
ii) Three-address code
iii) Triples
iv) DAG
v) Quadruples

a) Code generation and optimization: 5 5

Consider the following C codes and answer the questions (i) to (iii):

```
void bubbleSort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1])
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
        }
    }
```

i) Translate the C programming code into three-address code.
ii) Identify the basic block in three-address code.
Construct the flow graph from the three-address code.

b) Code optimization: 5 5

i) What are the structure preserving transformation of basic blocks? Explain one of them with appropriate example for code optimization.

ii) Construct the DAG for the following basic block.

$w := x \cdot y$
$x := w - z$
$y := x \cdot y$
$z := w - z$